带一个左半平面零点 $s=-z$ 的二阶系

输入阶跃 $U_{(s)} = \frac{1}{s}$

输出 $C_{(s)} = U_{(s)} \cdot G_{0(s)}$

$= \frac{1}{s} G_{0(s)}$

$t \to \infty$ 稳态

用终值定理

$$\lim_{s\to 0} s C_{(s)} = \lim_{s\to 0} s \cdot \frac{1}{s} G_{0(s)} = \lim_{s\to 0} G_{0(s)}$$

$$G_{0(s)} = \frac{\omega_n^2}{s^2+2\zeta\omega_n s+\omega_n^2} \quad \lim_{s\to 0} G_{(s)} = 1$$

引入零点 $s=-z$

若不除以 z，$\frac{\omega_n^2(s+z)}{s^2+2\zeta\omega_n s+\omega_n^2}$ $\lim_{s\to 0} G_{(s)} = z$

所以改成 $\frac{s+z}{z} \cdot \frac{\omega_n^2}{s^2+2\zeta\omega_n s+\omega_n^2}$ $\lim_{s\to 0} G_{(s)} = 1$

老

~~$PD_{(s)} = K_p + K_d \cdot s$~~ $= 1 + \frac{1}{z} \cdot s$

$L^{-1}[C(s)] = L^{-1}[(1+\frac{1}{z}\cdot s)\,C_0(s)]$

$= L^{-1}[(1+\frac{1}{z}s)\cdot\frac{1}{s}\cdot\frac{\omega_n^2}{s^2+2\xi\omega_n s+\omega_n^2}]$

不必关注 $C_0(s)$ 什么形式，看作整体

$C(s) = (1+\frac{1}{z}\cdot s)\cdot C_0(s)$

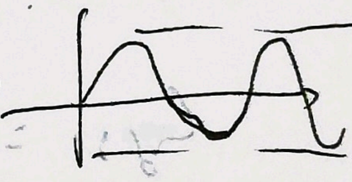
L^{-1}

$c(t) = c_0(t)+\frac{1}{z}\cdot\dot{c}_0(t)$

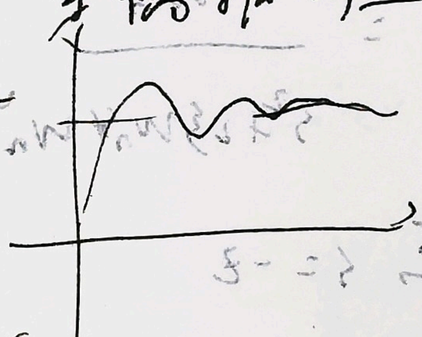
之前求了二阶单位阶跃响应

特例

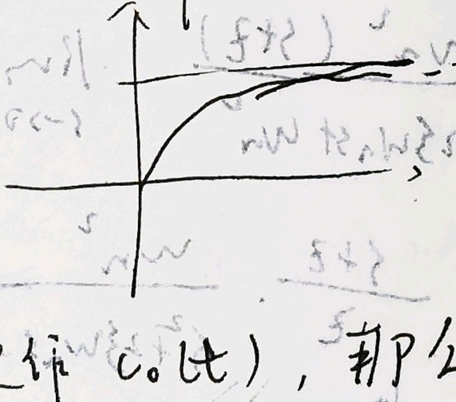
$\xi=0$ 无阻尼 等幅振荡



$0<\xi<1$ 欠阻尼

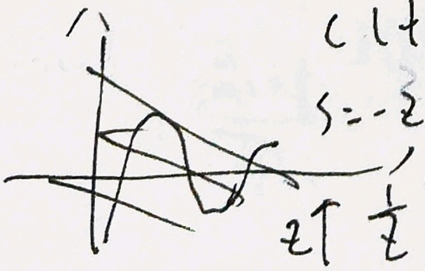


$\xi\geq1$



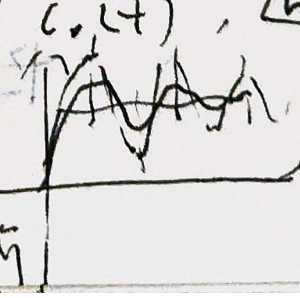
这些是 $c_0(t)$，那么上升段 $\dot{c}_0(t)>0$

$c(t)$ 就相当于叠加一个 $\dot{c}_0(t)$，超调变大



$s=-z$ 越靠近虚轴越明显

$z\uparrow$ $\frac{1}{z}\downarrow$ $z\to\infty$ $\frac{1}{z}=0$ 普通二阶



注：拐点不同

补、过 一下

令 $\zeta=0.3$

$$\frac{\omega_n^2}{s^2+2\zeta\omega_n s+\omega_n^2}$$

$$\frac{1}{s^2+0.6s+1}$$

~~引入 $z=1 \to z$.~~

$D.\ \frac{N}{1+N\frac{1}{s}}$

$\frac{s+z}{z}$ $\quad s+z=0$

$z=1,\ 2,\ 6$

零点 $-1,\ -2,\ -6$

$\frac{100}{1+100\frac{1}{s}}$

$=5$

$\overset{\varphi}{\boxed{1+\frac{1}{z}s}}$

直接代入就好

$$\frac{s+6}{6(s^2+0.6s+1)}$$

[1, 1] [1, 2] [1, 6]

[1, 0.6, 1] [2, 1.2, 2] [6, 3.6, 6]